

Question ID: 983412ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square prism has a height of **14** units. The volume of the prism is **2,016** cubic units. What is the length, in units, of an edge of the base?

Correct Answer: 12

Rationale

The correct answer is **12**. The volume, V , of a right square prism can be calculated using the formula $V = s^2h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is **2,016** cubic units and the height is **14** units. Substituting **2,016** for V and **14** for h in the formula $V = s^2h$ yields $2,016 = (s^2)(14)$. Dividing both sides of this equation by **14** yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is **12**.